

RFC-ATF-5

OMNIX Open Standard Series -- Fifth RFC

Agent Trust Fabric

Cognitive Governance Layer



OMNIX QUANTUM

Counterfactual Governance Engine | Grand Unified Governance Theory |
Temporal Governance Bridge

Document: RFC-ATF-5 -- OMNIX QUANTUM Open Standard

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RFC-ATF-2 (DOI 10.5281/zenodo.20241344)

RFC-ATF-3 (DOI 10.5281/zenodo.20247342)

RFC-ATF-4 (DOI 10.5281/zenodo.20368895)

Cryptography: Dilithium-3 ML-DSA-65 (NIST FIPS 204)

Formal Methods: Z3 SMT + TLA+ dual methodology

New Invariants: 18 -- CGE-INV-001-007, GUGT-INV-001-006, TGB-INV-001-005

Total ATF Inv.: 88 across 17 protocol families

New DB Tables: 6 -- counterfactual_forks, counterfactual_tokens, universal_invariant_receipts,
temporal_context_snapshots, regulatory_alignment_receipts, temporal_migration_records

Designation: ATF-CGL-Compliant -- fifth and highest compliance tier

Status: DRAFT -- pending Zenodo submission

Abstract

RFC-ATF-5 specifies the **Cognitive Governance Layer (CGL)** -- the fifth RFC in the OMNIX Agent Trust Fabric Open Standard series, extending RFC-ATF-1 through RFC-ATF-4. The CGL formally closes three governance gaps that no existing AI governance framework -- academic, regulatory, or commercial -- has previously addressed.

(1) The Decision Space Gap (Gap_DS). Prior ATF RFCs record the selected governance path with cryptographic certainty. They do not record the *decision space* -- the full set of governance outcomes reachable under parametric variation. The **Counterfactual Governance Engine (CGE)** computes M cryptographically sealed alternative paths (Counterfactual Fork Records, CFRs) at each evaluation, assembled into a Counterfactual Attestation Token (CAT) bound to the primary receipt. The CAT provides: offline-verifiable decision space evidence satisfying EU AI Act Art. 9 "alternatives considered" with PQC-signed artifacts; a **fragility_score** in [0.0, 1.0] quantifying decision robustness -- the first governance decision fragility metric in any published specification; and deterministic Variation Vectors ensuring independent reproducibility.

(2) The Universal Completeness Gap (Gap_UC). The **Grand Unified Governance Theory (GUGT)** derives six Universal Governance Invariants (UGI-001-006) by formal intersection of EU AI Act, NIST AI RMF, GCC/DIFC AI Regulation 2024, ISO/IEC 42001:2023, and UK AISI Evaluation Framework simultaneously. ATF-compliant systems satisfy all six UGIs by construction, earning GUGT-L3+ATF certification via a single PQC-signed Universal Invariant Receipt (UIR). This eliminates per-jurisdiction custom compliance analysis for ATF-compliant deployments.

(3) The Temporal Semantic Gap (Gap_TS). EU AI Act Art. 72 mandates 7-year retention for high-risk AI. The **Temporal Governance Bridge (TGB)** closes the interpretability gap between nanosecond-precision runtime governance and decade-scale regulatory review. A Temporal Context Snapshot (TCS) is embedded at nanosecond precision at the moment of every record issuance, capturing the complete regulatory and threshold context. A Regulatory Alignment Receipt (RAR) projects historical records to current frameworks at review time without modifying the original evidence (TGB-INV-002 non-destruction invariant).

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1. Introduction and Motivation

1.1 The ATF Open Standard Series

The OMNIX Agent Trust Fabric is an open standard series for cryptographic authority governance of autonomous AI agents. Each RFC addresses a distinct, formally scoped gap in the governance infrastructure required for enterprise and regulated AI deployments.

RFC	Layer	Defining Question	Status
RFC-ATF-1	Identity & Delegation	Who authorized this agent, with what authority, and can that be proved offline?	Published DOI: 10.5281/zenodo.20155016
RFC-ATF-2	Runtime Continuity	Was authority continuously valid throughout execution, and was HALT propagated correctly?	Published DOI: 10.5281/zenodo.20241344
RFC-ATF-3	Evidence & Forensic	Where does evidence go and who can verify it years later without any platform access?	Published DOI: 10.5281/zenodo.20247342
RFC-ATF-4	Proactive Governance	What happened between requests? Is recalibration safe? Are receipts semantically portable cross-domain?	Published DOI: 10.5281/zenodo.20368895
RFC-ATF-5	Cognitive Governance	What else could have happened? Is governance universally complete? Will evidence remain interpretable across time?	This document

1.2 Gap Analysis: What RFC-ATF-1/2/3/4 Left Open

The four prior RFCs collectively establish a complete, formally specified governance pipeline: identity and delegation (RFC-ATF-1), runtime integrity and HALT propagation (RFC-ATF-2), evidence lifecycle and forensic packages (RFC-ATF-3), and proactive veto plus semantic portability (RFC-ATF-4). This pipeline closes the execution governance gap.

Three structural questions remain outside the scope of Layers 1-4. These are not implementation gaps -- they are specification gaps that require new record types, new invariants, and new formal proofs. RFC-ATF-5 closes all three.

Q: What other governance outcomes were possible at decision time, and can any party verify this offline?

A: Gap_DS -- closed by CGE

Q: Does a single governance artifact certify compliance with all major regulatory frameworks simultaneously?

A: Gap_UC -- closed by GUGT

Q: Will a governance record issued today remain interpretable in 2031 under an amended EU AI Act?

A: Gap_TS -- closed by TGB

1.3 Scope of RFC-ATF-5

RFC-ATF-5 specifies: (1) the Counterfactual Governance Engine (CGE) -- new DB tables, record types CFR and CAT, 7 invariants; (2) the Grand Unified Governance Theory (GUGT) -- UGI-001-006, UIR record type, 4 conformance levels, 6 invariants; (3) the Temporal Governance Bridge (TGB) -- TCS, RAR, and TMR record types, projection rulebook, 5 invariants. Total: 18 new invariants, 6 new DB tables, 5 new record types.

All three modules are independently operable (CGL-INDEPENDENT-001). Failure of any CGL module is non-blocking for the primary ATF record. Each module degrades gracefully with flag notation on the primary record rather than halting the governance pipeline.

2. Problem Statement: The Cognitive Governance Gap

2.1 The Decision Space Gap (Gap_DS)

The ATF stack through RFC-ATF-4 guarantees that DP = SP (Documented Path = Selected Path) with cryptographic certainty via PQC signatures and append-only receipts. What remains undocumented is the Decision Space (DS) -- the full set of governance outcomes that existed at evaluation time under parametric variations of the authority configuration.

```
DECISION SPACE STRUCTURE:

DS = { NOMINAL (CES >= 80), MONITORING (CES 50-79), WARNING (CES 30-49), HALT (CES < 30) }

For a given evaluation with CES = 82.5 (primary outcome = NOMINAL):

+-----+-----+-----+-----+
| Outcome | Reachability | Documenter (pre RFC-ATF-5) |
+-----+-----+-----+-----+
| NOMINAL | Selected -- SP = DP | Primary receipt (PQC-signed) |
| MONITORING | Reachable at -20% VV | NOT documented |
| WARNING | Reachable at -15% budget VV | NOT documented |
| HALT | Reachable at -40% VV | NOT documented |
+-----+-----+-----+-----+

Gap_DS = DS \ {SP} = { MONITORING, WARNING, HALT }

Stakeholders who need Gap_DS documented:
[A] Regulatory auditors: EU AI Act Art. 9 -- "alternatives considered"
[B] Risk officers: fragility_score = |divergent_paths| / |total_paths|
[C] Legal counsel: "could the system have reached a HALT outcome?"

CGE closes Gap_DS: each element of Gap_DS becomes a PQC-signed CFR in the CAT.
```

2.2 The Universal Completeness Gap (Gap_UC)

The ATF stack satisfies Governance Completeness (GC) for specific (Framework, Agent-type) pairs. It does not formally assert Multi-Frame Completeness (MFC). Enterprise buyers deploying across EU + US + GCC + ISO jurisdictions must engage external counsel to verify compliance in each framework independently -- a significant cost barrier for adoption.

```
WITHOUT RFC-ATF-5 GUGT:

+-----+-----+-----+-----+
| EU AI Act | | NIST AI RMF | | GCC/DIFC | | ISO/IEC 42001 |
| compliance check | | compliance check | | compliance check | | compliance check |
| (custom mapping) | | (custom mapping) | | (custom mapping) | | (custom mapping) |
+-----+-----+-----+-----+

4 separate analyses, 4 separate artifacts, no unified proof

WITH RFC-ATF-5 GUGT:
```

```

+-----+
| Universal Invariant Receipt (UIR) GUGT-L3+ATF |
| UGI-001 [OK] UGI-002 [OK] UGI-003 [OK] |
| UGI-004 [OK] UGI-005 [OK] UGI-006 [OK] |
| PQC-signed ML-DSA-65 -- offline-verifiable |
+-----+

1 artifact satisfies EU + US + GCC + ISO + UK AISI simultaneously
ATF-compliant systems earn GUGT-L3+ATF by construction -- zero custom analysis

```

2.3 The Temporal Semantic Gap (Gap_TS)

EU AI Act Art. 72 mandates 7-year retention for high-risk AI decisions. A receipt issued in 2026 under "EU AI Act 2024 v1.0" with `nominal_threshold = 80.0` must remain interpretable when reviewed in 2033 under an amended framework -- without access to the original platform, without the issuing team, and without guessing what the regulatory context was.

This problem was formally identified during ATF Field Specification Partner Integration: *"The DSPP addresses semantic divergence at a point in time. The remaining open problem is the temporal dimension -- how does an auditor five years from now interpret a receipt issued under today's regulatory context, without the platform, without us, and without guessing?"* (Antonio Socorro, CAI-EXPERT-LAB)

```

WITHOUT TGB: WITH RFC-ATF-5 TGB:
+-----+ +-----+
| receipt.json | | receipt.json | |
| issued_at: 2026 | | +--> TCS (embedded at T=0 nanosecond) |
| outcome: NOMINAL | | | eu_ai_act_version: EU_AI_ACT_2024_v1.0 |
| ces_score: 82.5 | | | nominal_threshold: 80.0 |
| threshold: 80.0 | | | atf_spec_version: RFC-ATF-5_v1.0 |
| | | | tcs_seal: ML-DSA-65 [OK] |
| [no context] | | +--> RAR (produced at T=7yr, non-destr.) |
| [ambiguous in 2033] | | | projection to EU_AI_ACT_2033_v3.0 |
+-----+ | | original_record_integrity: VERIFIED |
Misclassification risk | | rar_seal: ML-DSA-65 [OK] |
+-----+
Zero ambiguity -- self-contained projection

```

2.4 Formal Definition: Gap_CG

Gap_CG = Gap_DS U Gap_UC U Gap_TS

Gap_CG is the Cognitive Governance Gap -- the set of all governance properties that an ATF-compliant system through RFC-ATF-4 does not formally specify. RFC-ATF-5 closes Gap_CG completely. The Cognitive Governance Layer (CGL) is the set of protocol elements, record types, invariants, and formal proofs that constitute this closure.

3. Architecture Overview: Cognitive Governance Layer

3.1 ATF Stack -- Five-Layer Architecture

```
+=====+
| ATF PROTOCOL STACK -- FIVE-LAYER COMPLETE ARCHITECTURE |
+=====+
| |
| Layer 5 -- COGNITIVE GOVERNANCE LAYER (RFC-ATF-5) ----- * THIS RFC|
| +-----+
| | CGE: Counterfactual Governance Engine |
| | -> M alternative governance paths, PQC-sealed as CAT |
| | -> fragility_score in [0.0, 1.0] -- decision robustness metric |
| | GUGT: Grand Unified Governance Theory |
| | -> 6 UGIs derived by 5-framework intersection |
| | -> UIR certifies EU+NIST+GCC+ISO+UK simultaneously |
| | TGB: Temporal Governance Bridge |
| | -> TCS at nanosecond issuance -- RAR at 7-year review |
| | -> TMR at each evidence lifecycle transition |
| +-----+
| |
| Layer 4 -- PROACTIVE GOVERNANCE PLANE (RFC-ATF-4) |
| +-----+
| | AGVP: Anticipatory Governance Veto Protocol |
| | SSD: Structural Shift Detection |
| | DSPP: Dynamic Semantic Portability Protocol |
| +-----+
| |
| Layer 3 -- EVIDENCE AND FORENSIC PLANE (RFC-ATF-3) |
| +-----+
| | GPIL: Governance Policy Interoperability Layer |
| | ELC: Evidence Lifecycle Classification (HOT/WARM/COLD) |
| | OEP: OMNIX Evidence Package -- offline-verifiable forensic bundle |
| | FVP: Forensic Verification Protocol |
| +-----+
| |
| Layer 2 -- RUNTIME CONTINUITY PLANE (RFC-ATF-2) |
| +-----+
| | RCR: Runtime Continuity Record |
| | CES: Coherence Evaluation Score (continuous health signal) |
| | AFG: Authority Fragmentation Guard |
| | RC: Reauthorization Challenge with HALT propagation |
```

```

| +-----+
| |
| Layer 1 -- IDENTITY AND DELEGATION PLANE (RFC-ATF-1) |
| +-----+
| | AIR: Agent Identity Record -- AID-{DOMAIN}-{16HEX} |
| | DR: Delegation Receipt -- authority budget chain |
| | TAR: Temporal Admissibility Record -- nanosecond admission proof |
| | Trust Lattice -- DAG of all delegation receipts |
| +-----+
| |
+=====+
| Invariant count: L1=6, L2=8, L3=40, L4=16, L5=18 --> TOTAL: 88 |
| Compliance tier: ATF-CGL-Compliant (5th and highest tier) |
+=====+

```

3.2 CGL Module Independence and Failure Isolation

All three CGL modules are independently operable. The failure of any module does not block issuance of the primary ATF record. This is enforced by the CGL Independence Principle:

Module	Dependency on Primary Record	Blocking on Failure	Flag on Failure
CGE	Primary record must be persisted FIRST (CGE-INV-002)	Non-blocking -- CAT is async	CGE_INCOMPLETE on primary record
GUGT	UIR assessed against deployment configuration, not per-record	Non-blocking -- on-demand per deployment	GUGT_UNASSESSED on deployment
TGB	TCS embedded synchronously, then bound to primary record	Non-blocking -- TCS failure retried async	TGB_INCOMPLETE on primary record

3.3 CGL Record Sequencing

```

For every primary ATF record R:

Step 1 -- Issue and persist R (Layer 1-4 record, all prior invariants apply)
R.content_hash sealed -- R.pqc_signature applied -- DB INSERT committed

Step 2 -- TCS embedding (SYNCHRONOUS, part of R creation pipeline)
TCS.regulatory_context captured at R.issued_at nanosecond
tcs_hash included in R.posture_state_hash
TCS persisted -- TCS PQC-signed

Step 3 -- CGE fork computation (ASYNCR, begins after Step 1 completes)
M CFRs computed with deterministic VVs (seed = SHA256(eval_id || receipt_id))
Each CFR: read-only re-evaluation, no receipts/HALTs produced
CAT assembled: cat_root_hash = sha256(sorted(cfr_content_hashes))
CAT PQC-signed and persisted

```

Step 4 -- UIR issuance (ON DEMAND, not per-record, per deployment)
GUGT assessment run against deployment configuration
UIR issued with 6 UGI results, PQC-signed

Invariant order guarantee:
 $T(\text{primary.persist}) < T(\text{tcs.persist}) < T(\text{cge.start})$ [CGE-INV-002 + TGB-INV-001]
Any CGL failure is non-blocking -- primary record always valid and immutable

4. Counterfactual Governance Engine (CGE)

The CGE computes M alternative governance paths at the moment of a primary ATF governance evaluation. M is configurable via `CGE_FORK_COUNT` (default: 3, range: 1-7). Each path applies a deterministic Variation Vector (VV) to the primary evaluation inputs and re-executes governance logic in read-only simulation mode. The CGE produces no HALTs, no escalations, and no side effects on the live system. Its sole output is a set of PQC-signed Counterfactual Fork Records (CFRs) assembled into a Counterfactual Attestation Token (CAT).

4.1 Decision Space Architecture

```
PRIMARY EVALUATION (sealed first -- CGE-INV-002):
+-----+
| ATFDR-4A2B8F1C3D5E7A9B |
| outcome: NOMINAL ces_score: 82.5 pqc_sig: ML-DSA-65 [OK] |
+-----+
| DB INSERT committed
v CGE begins NOW (async)
+-----+
| VARIATION VECTOR GENERATION |
| seed = SHA256(evaluation_id || primary_receipt_id) |
| VV-1: authority_budget_delta_pct = -0.20 |
| VV-2: ces_threshold_nominal_override = 88.0 |
| VV-3: delegation_depth_limit = 3 |
| (Deterministic: same seed always produces same VVs) |
+-----+
| | |
v v v (read-only re-evaluations)
+-----+ +-----+ +-----+
| CFR-1 | | CFR-2 | | CFR-3 |
| MONITORING | | NOMINAL | | HALT |
| ces: 72.0 | | ces: 82.5 | | ces: 15.3 |
| diverges: | | diverges: | | diverges: |
| true | | false | | true |
| ML-DSA-65 | | ML-DSA-65 | | ML-DSA-65 |
+-----+ +-----+ +-----+
| | |
+-----+
|
+-----+
| COUNTERFACTUAL ATTESTATION TOKEN (CAT) |
| cat_root_hash = sha256(sort([cfr1_hash, cfr2_hash, cfr3_hash])) |
| cfr_count: 3 divergence_count: 2 |
```

```
| fragility_score: 0.67 (2 out of 3 paths diverge) |
| cat_seal: ML-Dsa-65 over cat_root_hash |
| verification: offline, no platform access required |
+-----+
```

4.2 Variation Vector Design and Determinism

Variation Vectors are the only mechanism by which CGE explores the decision space. Every VV is deterministically derived from the evaluation seed, ensuring that any independent party can reproduce all CFRs given only the primary receipt and the CGE configuration. This is the basis for CGE-INV-007 (offline verifiability).

VV Field Category	Description	Bound (CGE-INV-005)	Example Delta
authority_budget_delta_pct	Percentage change applied to authority_budget_granted	$\text{abs}(\text{delta}) \leq 0.40$	-0.20 (20% reduction)
ces_threshold_nominal_override	Override of the NOMINAL outcome threshold	$\text{abs}(\text{delta}) \leq 0.40$	88.0 (raised from 80.0)
delegation_depth_limit	Maximum delegation chain depth allowed	$\text{abs}(\text{delta}) \leq 0.40$	3 (reduced from 5)
monitoring_window_override	Temporal window for monitoring aggregation	$\text{abs}(\text{delta}) \leq 0.40$	300s (reduced from 600s)
fragmentation_limit_override	AFG fragmentation threshold	$\text{abs}(\text{delta}) \leq 0.40$	0.75 (from 0.90)

VVs exceeding the `CGE_MAX_VARIATION_PCT` bound are rejected with a null CFR carrying flag `REJECTED`. Null CFRs do NOT count toward `cfr_count` for CGE-INV-001 purposes. This prevents exploration of parametric spaces so extreme they have no governance interpretive value.

4.3 Counterfactual Fork Record (CFR) Specification

```
CFR (Counterfactual Fork Record) -- identifier: CFR-{16HEX}

COMMITTED FIELDS (included in posture_state_hash_cf):
cfr_id : "CFR-4A2B8F1C3D5E7A9B"
primary_receipt_id : "ATFDR-..." (binding to primary record)
evaluation_id : "CGE-EVAL-..." (evaluation session)
variation_vector : { "authority_budget_delta_pct": -0.20, ... }
counterfactual_ces_score : 72.0
counterfactual_outcome : "MONITORING"
diverges_from_primary : true
issued_at : "2026-05-26T14:00:00.123456789+00:00"

SEAL FIELDS:
posture_state_hash_cf : SHA256(canonical JSON of committed fields)
content_hash_cf : SHA256(canonical JSON of all fields incl. posture_hash)
pgc_algorithm : "ML-Dsa-65"
```

```

pgc_signature : base64(ML-DSA-65.sign(content_hash_cf))

METADATA:
cfr_status : "ACCEPTED" | "REJECTED" (VV bound violation)
vv_seed : SHA256(evaluation_id || primary_receipt_id)
rejection_reason : null | "CGE_VV_BOUND_VIOLATION"
cat_id : "CAT-..." (parent token)

```

4.4 Counterfactual Attestation Token (CAT) Specification

```

CAT (Counterfactual Attestation Token) -- identifier: CAT-{16HEX}

COMMITTED FIELDS:
cat_id : "CAT-9F3A7B2C1D4E5F6A"
primary_receipt_id : "ATFDR-..." (binding to primary record)
evaluation_id : "CGE-EVAL-..."
cfr_ids : ["CFR-...", "CFR-...", "CFR-..."]
cfr_count : 3
divergence_count : 2
cat_root_hash : SHA256(sort([cfr_1_content_hash, cfr_2_content_hash, ...]))
fragility_score : 0.6667 (divergence_count / cfr_count)
cge_fork_count_config : 3
issued_at : "2026-05-26T14:00:00.200000000+00:00"

SEAL FIELDS:
content_hash_cat : SHA256(canonical JSON of all fields excl. seal fields)
pgc_algorithm : "ML-DSA-65"
cat_seal : base64(ML-DSA-65.sign(content_hash_cat))

COMPUTATION:
cat_root_hash = sha256("|".join(sorted([cfr.content_hash_cf for cfr in cfrs])))
fragility_score = divergence_count / cfr_count in [0.0, 1.0]

```

4.5 fragility_score: Decision Robustness Metric

The `fragility_score` is the first formally specified decision robustness metric in any published AI governance standard. It quantifies what proportion of the explored decision space diverges from the primary governance outcome under parametric variation.

fragility_score	Interpretation	Governance Signal	Recommended Action
0.0	All counterfactuals agree with primary outcome	Decision robust -- parameter-insensitive	No action required
0.01 - 0.33	Minority of paths diverge	Decision moderately robust	Periodic review recommended
0.34 - 0.66	Roughly half of paths diverge	Decision sensitive to parameters	Risk review required

fragility_score	Interpretation	Governance Signal	Recommended Action
0.67 - 0.99	Majority of paths diverge	Decision fragile -- high parameter sensitivity	Escalate to human review
1.0	All counterfactuals diverge from primary	Maximum fragility -- fully parameter-dependent	Mandatory human override

```

FRAGILITY SCORE COMPUTATION (CGE-INV-004):
fragility_score = divergence_count / cfr_count

Where:
divergence_count = count(cfr for cfr in cfrs if cfr.diverges_from_primary == true)
cfr_count = count(cfr for cfr in cfrs if cfr.cfr_status == "ACCEPTED")

CONSTRAINTS:
fragility_score in [0.0, 1.0] -- Z3 proof target CGE-FRAGILITY-BOUND-LO/HI
fragility_score > 1.0 is rejected as a computation error (CGE-INV-004 violation)
fragility_score is bound to CAT via content_hash_cat -- tamper-evident

```

4.6 Offline Verification Protocol

Any party can verify a CAT and its CFRs without access to the issuing platform, without API credentials, and without network access. The following procedure is sufficient:

```

OFFLINE CAT VERIFICATION PROCEDURE (CGE-INV-007):

Input: primary_receipt.json, cat.json, cfr_list.json, public_key.pem

Step 1 -- Verify primary receipt integrity
hash_check = sha256(canonical_json(primary_receipt.committed_fields))
assert hash_check == primary_receipt.content_hash
verify ML-DSA-65(primary_receipt.pqc_signature, primary_receipt.content_hash, pubkey)

Step 2 -- Verify CAT integrity
hash_check = sha256(canonical_json(cat.committed_fields))
assert hash_check == cat.content_hash_cat
verify ML-DSA-65(cat.cat_seal, cat.content_hash_cat, pubkey)

Step 3 -- Verify CAT root hash
computed_root = sha256("".join(sorted([cfr.content_hash_cf for cfr in cfrs])))
assert computed_root == cat.cat_root_hash

Step 4 -- Verify each CFR
For each cfr in cfr_list:
hash_check = sha256(canonical_json(cfr.committed_fields))
assert hash_check == cfr.content_hash_cf
verify ML-DSA-65(cfr.pqc_signature, cfr.content_hash_cf, pubkey)
assert cfr.primary_receipt_id == primary_receipt.receipt_id

Step 5 -- Verify VV determinism

```

```

for each cfr in cfr_list:
    expected_seed = sha256(cfr.evaluation_id + "|" + cfr.primary_receipt_id)
    assert cfr.vv_seed == expected_seed

Step 6 -- Verify fragility_score
computed_fragility = count(cfr for cfr in cfrs if cfr.diverges_from_primary) / len(cfrs)
assert abs(computed_fragility - cat.fragility_score) < 0.0001

Result: VERIFIED | NOT_VERIFIED (with specific failure step)

```

4.7 CGE Invariants: CGE-INV-001-007

Invariant	Statement	Violation Consequence
CGE-INV-001	For every evaluation with CGE_ENABLED=true, a CAT with at least 1 accepted CFR MUST be produced and persisted	CGE_INCOMPLETE flag written to primary record; CAT retried async
CGE-INV-002	Primary decision outcome MUST be identical regardless of whether CGE_ENABLED is true or false	Critical structural defect -- CGE is strictly read-only post-primary-commit
CGE-INV-003	Every accepted CFR MUST carry a valid ML-DSA-65 signature over its content_hash_cf	Unsigned or invalid-signature CFR rejected from CAT; logged as CGE_CFR_SIG_FAIL
CGE-INV-004	cat_root_hash = sha256(sorted(cfr_content_hashes)); fragility_score = divergence_count / cfr_count in [0.0, 1.0]	Root hash mismatch or out-of-range fragility_score treated as computation error
CGE-INV-005	abs(VV field delta) <= CGE_MAX_VARIATION_PCT (default 0.40, max 0.40)	VV rejected; null CFR with REJECTED status; logged CGE_VV_BOUND_VIOLATION
CGE-INV-006	Tables atf_counterfactual_forks and atf_counterfactual_tokens are append-only; UPDATE and DELETE are prohibited	DB trigger rejects update/delete; event logged as CGE_IMMUTABILITY_VIOLATION
CGE-INV-007	A CAT MUST be independently verifiable offline using only: primary receipt, cat.json, cfr_list.json, public_key.pem	Verification procedure failure indicates structural defect in CAT construction

4.8 CGE Database Schema

```

-- Table: atf_counterfactual_forks (append-only, CGE-INV-006)
CREATE TABLE IF NOT EXISTS atf_counterfactual_forks (
    id SERIAL PRIMARY KEY,
    cfr_id VARCHAR(50) UNIQUE NOT NULL,
    cat_id VARCHAR(50) NOT NULL,
    primary_receipt_id VARCHAR(100) NOT NULL,
    evaluation_id VARCHAR(100) NOT NULL,
    variation_vector JSONB NOT NULL,
    counterfactual_ces_score DECIMAL(10,4),
    counterfactual_outcome VARCHAR(30),

```



```

diverges_from_primary BOOLEAN NOT NULL DEFAULT false,
cfr_status VARCHAR(20) NOT NULL DEFAULT 'ACCEPTED',
rejection_reason VARCHAR(100),
vv_seed VARCHAR(64),
posture_state_hash_cf VARCHAR(64) NOT NULL,
content_hash_cf VARCHAR(64) NOT NULL,
pqc_algorithm VARCHAR(30) DEFAULT 'ML-DSA-65',
pqc_signature TEXT,
issued_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- Table: atf_counterfactual_tokens (append-only, CGE-INV-006)
CREATE TABLE IF NOT EXISTS atf_counterfactual_tokens (
id SERIAL PRIMARY KEY,
cat_id VARCHAR(50) UNIQUE NOT NULL,
primary_receipt_id VARCHAR(100) NOT NULL,
evaluation_id VARCHAR(100) NOT NULL,
cfr_ids JSONB NOT NULL,
cfr_count INTEGER NOT NULL,
divergence_count INTEGER NOT NULL DEFAULT 0,
cat_root_hash VARCHAR(64) NOT NULL,
fragility_score DECIMAL(8,4) NOT NULL
CHECK (fragility_score >= 0.0 AND fragility_score <= 1.0),
cge_fork_count_config INTEGER NOT NULL DEFAULT 3,
content_hash_cat VARCHAR(64) NOT NULL,
pqc_algorithm VARCHAR(30) DEFAULT 'ML-DSA-65',
cat_seal TEXT,
issued_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

```

5. Grand Unified Governance Theory (GUGT)

The GUGT establishes a formal meta-layer above the ATF protocol stack. It derives six Universal Governance Invariants (UGI-001-006) by exhaustive intersection analysis of five major regulatory frameworks, establishing the minimal sufficient set of properties for Multi-Frame Completeness (MFC). The GUGT does not replace any prior ATF invariant -- it provides a unified certification artifact (UIR) that proves all six UGIs are satisfied simultaneously, for any agent type, under any covered framework.

5.1 Universal Invariant Architecture

The GUGT invariant derivation follows a four-step methodology: (1) extract governance requirements from each framework at the clause level; (2) map each clause to ATF protocol elements; (3) identify requirements present in all five frameworks simultaneously (the intersection); (4) formalize each intersecting requirement as a UGI with a precise ATF mechanism reference.

```
FIVE-FRAMEWORK INTERSECTION METHODOLOGY:

Framework | Human | Offline | Exec | Pre-Comm | No-Self | Self- |
| Anchor | Verify | Bound | Posture | Modify | Contain |
-----+-----+-----+-----+-----+-----+-----+
EU AI Act | Art.14 | Art.11 | Art.9 | Art.9(7) | Art.14 | Art.18 |
NIST AI RMF | GOVERN1.1 | MAP 5.2 | MANAGE2.2 | MEASURE2.5 | GOVERN6.1 | MAP 5.1 |
GCC/DIFC | Art.8 | Art.12 | Art.9 | Art.9 | Art.10 | Art.14 |
ISO/IEC 42001 | Sec.6.2 | Sec.9.1 | Sec.8.4 | Sec.8.5 | Sec.6.1.2 | Sec.9.3 |
UK AISI | Sec.3 | Sec.4 | Sec.4 | Sec.4 | Sec.5 | Sec.5 |
-----+-----+-----+-----+-----+-----+-----+
INTERSECTION | UGI-001 | UGI-002 | UGI-003 | UGI-004 | UGI-005 | UGI-006 |
-----+-----+-----+-----+-----+-----+-----+

All five frameworks require all six properties simultaneously.
ATF-compliant systems satisfy all six by construction -> GUGT-L3+ATF by construction.
```

5.2 Framework Intersection Methodology

The intersection was conducted at clause level, not at framework level. Each of the five frameworks was mapped to a structured requirement matrix. A requirement is included in the UGI set only if an equivalent requirement exists in all five frameworks -- weaker requirements that appear in only four or fewer frameworks are not universalized.

The intersection yields exactly six requirements. This is not a design choice -- it is the mathematical result of the cross-framework analysis. No requirements were added or removed based on ATF capability. The six UGIs represent the complete intersection of the five frameworks over the governance domain.

5.3 Universal Governance Invariants UGI-001-006

UGI-001 -- Human Authority Anchor

Statement: Every governance decision by an autonomous AI agent MUST trace to an identified, cryptographically bound human principal via a verifiable delegation chain, provable offline.

ATF Mechanism: ATF DR chain_root_id -> Tier-1 human principal. ATF-INV-002 (acyclicity) + ATF-INV-006 (offline verifiability) satisfy UGI-001 by construction.

Framework Clauses: EU Art.14/17, NIST GOVERN 1.1, GCC Art.8, ISO Sec.6.2, UK AISI Sec.3

UGI-002 -- Offline-Verifiable Decision Evidence

Statement: Every governance decision record MUST be independently verifiable by any authorized party using only the artifacts themselves, without accessing the issuing platform or any network service.

ATF Mechanism: OEP two-phase PQC seal (OEP-INV-001 through OEP-INV-006). FEA-INV-001 through FEA-INV-005. CLI verifier.

Framework Clauses: EU Art.11/12, NIST MAP 5.2, GCC Art.12, ISO Sec.9.1, UK AISI Sec.4

UGI-003 -- Execution-Time Boundary Enforcement

Statement: Governance controls MUST be enforced at the moment of execution, not applied retrospectively, with a cryptographically sealed artifact produced at the exact moment of enforcement.

ATF Mechanism: TAR (Temporal Admissibility Record) issued BEFORE governance logic executes. RCR HALT at nanosecond precision (RGC-INV-003, RGC-INV-005).

Framework Clauses: EU Art.9, NIST MANAGE 2.2, GCC Art.9, ISO Sec.8.4, UK AISI Sec.4

UGI-004 -- Pre-Committed Posture Assessment

Statement: The input state assessment MUST be cryptographically committed before the governance output is computed, preventing post-hoc rationalization of the decision basis.

ATF Mechanism: posture_state_hash computed over committed input fields BEFORE content_hash computation (ATF-INV-003). Pre-computation ordering enforced in pipeline.

Framework Clauses: EU Art.9(7), NIST MEASURE 2.5, GCC Art.9, ISO Sec.8.5, UK AISI Sec.4

UGI-005 -- Self-Modification Prohibition

Statement: AI agent authority parameters MUST NOT be self-modifiable above defined thresholds. All modifications MUST be bounded and produce verifiable audit artifacts.

ATF Mechanism: Auto-Modification Guard (AMG, ADR-144): maximum 10% per event, 30% cumulative. AGVP watchdog for anticipatory veto of modification attempts.

Framework Clauses: EU Art.14(4), NIST GOVERN 6.1, GCC Art.10, ISO Sec.6.1.2, UK AISI Sec.5

UGI-006 -- Self-Contained Evidence Reconstruction

Statement: The complete governance evidence chain MUST be reconstructable from the receipt artifacts alone, without access to any external system, service, or database.

ATF Mechanism: OEP package contains: DR -> TAR -> RCR -> GovernanceReceipt with embedded delegator public key (ADR-165). Full chain reconstructable from OEP alone.

Framework Clauses: EU Art.18/72, NIST MAP 5.1, GCC Art.14, ISO Sec.9.3, UK AISI Sec.5

5.4 Universal Invariant Receipt (UIR) Specification

```
UIR (Universal Invariant Receipt) -- identifier: UIR-{16HEX}

COMMITTED FIELDS:
uir_id : "UIR-7F3A2B1C4D5E6F7A"
assessed_protocol : "ATF" (or other governance protocol identifier)
atf_spec_version : "RFC-ATF-5_v1.0"
agent_type_coverage : ["LLM", "FINANCIAL", "MEDICAL", "ROBOTIC"]
ugi_001_status : "PASS" | "PARTIAL" | "FAIL"
ugi_001_evidence_ref : "ATFDR-..." (binding to supporting ATF record)
ugi_002_status : "PASS"
ugi_002_evidence_ref : "OEP-..."
ugi_003_status : "PASS"
ugi_003_evidence_ref : "ATFTAR-..."
ugi_004_status : "PASS"
ugi_004_evidence_ref : "ATFDR-..." (posture_state_hash verification)
ugi_005_status : "PASS"
ugi_005_evidence_ref : "AMG-AUDIT-..."
ugi_006_status : "PASS"
ugi_006_evidence_ref : "OEP-..."
overall_gugt_status : "GUGT_COMPLIANT" (all 6 PASS)
| "GUGT_PARTIAL" (at least one PARTIAL, no FAIL)
| "GUGT_NON_COMPLIANT" (at least one FAIL)
conformance_level : "GUGT-L3+ATF"
framework_coverage : ["EU_AI_ACT_ART14", "NIST_GOVERN_1.1", ...]
issued_at : "2026-05-26T14:00:00.000000000+00:00"

SEAL FIELDS:
content_hash_uir : SHA256(canonical JSON of committed fields)
pqc_algorithm : "ML-DSA-65"
uir_seal : base64(ML-DSA-65.sign(content_hash_uir))
```

5.5 GUGT Conformance Levels

Level	UGIs Required	Additional Requirements	Typical Deployer
GUGT-L1 Basic	UGI-001 + UGI-002	Human authority anchor + offline evidence verification	Any system with identity chain and signed receipts
GUGT-L2 Runtime	GUGT-L1 + UGI-003 + UGI-004	Execution-time boundary enforcement + pre-committed posture	Runtime-enforcement governance systems
GUGT-L3 Full	GUGT-L2 + UGI-005 + UGI-006	Self-modification prohibition + self-contained evidence reconstruction	Full-stack governance systems (non-ATF)
GUGT-L3+ATF	GUGT-L3 + full ATF stack	All 88 ATF invariants (RFC-ATF-1 through RFC-ATF-5); ATF-CGL-Compliant status	ATF-CGL-Compliant systems -- earned by construction

GUGT conformance levels are strictly hierarchical: GUGT-L3+ATF implies GUGT-L3, GUGT-L3 implies GUGT-L2, GUGT-L2 implies GUGT-L1. This monotonicity is enforced by GUGT-INV-006 and verified by Z3 proof target GUGT-UGI-MONOTONICITY.

5.6 Cross-Jurisdiction Framework Mapping

UGI	EU AI Act	NIST AI RMF	GCC/DIFC 2024	ISO/IEC 42001	UK AISI
UGI-001	Art. 14, 17	GOVERN 1.1	Art. 8	Sec. 6.2	Sec. 3
UGI-002	Art. 11, 12	MAP 5.2	Art. 12	Sec. 9.1	Sec. 4
UGI-003	Art. 9	MANAGE 2.2	Art. 9	Sec. 8.4	Sec. 4
UGI-004	Art. 9(7)	MEASURE 2.5	Art. 9	Sec. 8.5	Sec. 4
UGI-005	Art. 14(4)	GOVERN 6.1	Art. 10	Sec. 6.1.2	Sec. 5
UGI-006	Art. 18, 72	MAP 5.1	Art. 14	Sec. 9.3	Sec. 5

5.7 GUGT Invariants: GUGT-INV-001-006

Invariant	Statement	Enforcement
GUGT-INV-001	UIR MUST assess all 6 UGIs; overall_gugt_status MUST reflect the minimum across all UGI statuses	UIR with fewer than 6 UGI assessments is structurally invalid and rejected
GUGT-INV-002	Framework clause references in UIR.framework_coverage MUST be specific (FRAMEWORK_CLAUSE format, not generic framework name)	Generic references (e.g., "EU AI Act" without article) rejected as non-specific
GUGT-INV-003	UIR MUST carry a valid ML-DSA-65 seal over content_hash_uir before persistence or distribution	Unsealed UIR rejected at DB insert level; cannot be distributed
GUGT-INV-004	UIR MUST explicitly declare agent_type_coverage; claimed coverage MUST match declared types	Claim-coverage mismatch logged as GUGT_COVERAGE_MISMATCH and UIR invalidated

Invariant	Statement	Enforcement
GUGT-INV-005	GUGT_COMPLIANT status MUST NOT relax any prior ATF invariant; all 88 invariants remain in full force	Any UIR that attempts to waive an ATF invariant is rejected as GUGT_WAIVER_VIOLATION
GUGT-INV-006	GUGT conformance levels are strictly hierarchical: GUGT-Lk implies GUGT-Lj for all $j < k$	Level claim failing monotonicity check rejected; Z3 proof target GUGT-UGI-MONOTONICITY

5.8 GUGT Database Schema

```
-- Table: gugt_universal_invariant_receipts
CREATE TABLE IF NOT EXISTS gugt_universal_invariant_receipts (
  id SERIAL PRIMARY KEY,
  uir_id VARCHAR(50) UNIQUE NOT NULL,
  assessed_protocol VARCHAR(50) NOT NULL DEFAULT 'ATF',
  atf_spec_version VARCHAR(30),
  agent_type_coverage JSONB NOT NULL,
  ugi_001_status VARCHAR(20) NOT NULL CHECK (ugi_001_status IN ('PASS','PARTIAL','FAIL')),
  ugi_001_evidence_ref VARCHAR(100),
  ugi_002_status VARCHAR(20) NOT NULL CHECK (ugi_002_status IN ('PASS','PARTIAL','FAIL')),
  ugi_002_evidence_ref VARCHAR(100),
  ugi_003_status VARCHAR(20) NOT NULL CHECK (ugi_003_status IN ('PASS','PARTIAL','FAIL')),
  ugi_003_evidence_ref VARCHAR(100),
  ugi_004_status VARCHAR(20) NOT NULL CHECK (ugi_004_status IN ('PASS','PARTIAL','FAIL')),
  ugi_004_evidence_ref VARCHAR(100),
  ugi_005_status VARCHAR(20) NOT NULL CHECK (ugi_005_status IN ('PASS','PARTIAL','FAIL')),
  ugi_005_evidence_ref VARCHAR(100),
  ugi_006_status VARCHAR(20) NOT NULL CHECK (ugi_006_status IN ('PASS','PARTIAL','FAIL')),
  ugi_006_evidence_ref VARCHAR(100),
  overall_gugt_status VARCHAR(30) NOT NULL,
  conformance_level VARCHAR(20) NOT NULL,
  framework_coverage JSONB NOT NULL,
  content_hash_uir VARCHAR(64) NOT NULL,
  pqc_algorithm VARCHAR(30) DEFAULT 'ML-DSA-65',
  uir_seal TEXT,
  issued_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

6. Temporal Governance Bridge (TGB)

AI governance operates simultaneously at two radically different time scales: nanoseconds (runtime enforcement) and years (regulatory review, EU AI Act Art. 72 mandates 7-year retention for high-risk AI). No prior AI governance specification provides a formal bridge between these scales. The TGB provides this bridge through three record types: the Temporal Context Snapshot (TCS), the Regulatory Alignment Receipt (RAR), and the Temporal Migration Record (TMR).

6.1 Two-Scale Architecture: Nanoseconds to Years

```
TGB TWO-SCALE ARCHITECTURE:

TIME AXIS:
----T=0ns-----T=6mo-----T=3yr-----T=7yr-----+
| | | | |
MICRO-SCALE LIFECYCLE LIFECYCLE MACRO-SCALE |
[nanoseconds] [months] [years] [7yr audit] |
| | | | |
+---v---+ +---v-----+ +---v-----+ +---v-----+ |
|Primary| |HOT -> WARM | |WARM -> COLD| |Regulatory | |
|record | |transition | |transition | |audit review | |
|issued | |TMR-001 | |TMR-002 | |RAR produced | |
+---+---+ +-----+ +-----+ +-----+ +-----+ |
| | |
v (sync, T+100ns) | |
+-----+ | |
| TCS embedded in primary record | | |
| eu_ai_act_version: "EU_AI_ACT_2024_v1.0" |<-+ |
| nist_ai_rmf_version: "NIST_AI_RMF_2023_v1.0" | |
| atf_spec_version: "RFC-ATF-5_v1.0" | |
| nominal_threshold: 80.0 | RAR reads |
| halt_threshold: 20.0 | TCS as |
| issued_at_ns: 1748268000123456789 | source |
| tcs_seal: ML-DSA-65 [OK] | |
+-----+ |
|
KEY PROPERTIES: |
- TCS never modified after issuance (TGB-INV-002) |
- RAR projects context -- does NOT touch primary record |
- RAR computable offline from: record + TCS + signed rulebook |
- TMR issued at each lifecycle transition (HOT->WARM, WARM->COLD) |
```

6.2 Temporal Context Snapshot (TCS) Specification

The TCS is the core innovation of the TGB. It is the first record type in any published governance specification designed explicitly to capture the interpretive context at nanosecond precision at the moment of record issuance. Without the TCS, a receipt is a fact without context. With the TCS, the receipt is a self-interpreting artifact.

```
TCS (Temporal Context Snapshot) -- identifier: TCS-{16HEX}

COMMITTED FIELDS:
tcs_id : "TCS-9B3F2A1D4C5E7F8A"
parent_record_id : "ATFDR-..." (the record this TCS is embedded in)
parent_record_type : "DR" | "TAR" | "RCR" | "GovernanceReceipt" | ...
issued_at_ns : 1748268000123456789 (nanosecond UNIX timestamp)
issued_at_iso : "2026-05-26T14:00:00.123456789+00:00"

regulatory_context:
eu_ai_act_version : "EU_AI_ACT_2024_v1.0"
nist_ai_rmf_version : "NIST_AI_RMF_2023_v1.0"
gcc_difc_version : "GCC_DIFC_AI_REG_2024_v1.0"
iso_42001_version : "ISO_42001_2023_v1.0"
atf_spec_version : "RFC-ATF-5_v1.0"
active_adr_count : 184

threshold_context:
nominal_threshold : 80.0
monitoring_lower : 50.0
warning_lower : 30.0
halt_threshold : 20.0
fragmentation_limit : 0.90
max_delegation_depth : 5

deployment_context:
agent_type : "LLM"
domain : "FINANCE"
jurisdiction : "EU"

SEAL FIELDS:
tcs_hash : SHA256(canonical JSON of committed fields)
pqc_algorithm : "ML-DSA-65"
tcs_seal : base64(ML-DSA-65.sign(tcs_hash))

BINDING TO PARENT RECORD:
tcs_hash is included in parent_record.posture_state_hash
Any tampering with TCS is detectable from parent record content_hash
```

6.3 Regulatory Alignment Receipt (RAR) Specification

The RAR is produced at review time (months or years after the original record issuance) to project a historical governance record to the current regulatory framework. The RAR is strictly non-destructive (TGB-INV-002): it

does not modify the source record, the source TCS, or any other original artifact. It adds interpretive projection alongside them.

```
RAR (Regulatory Alignment Receipt) -- identifier: RAR-{16HEX}

COMMITTED FIELDS:
rar_id : "RAR-2C4E6A8B1D3F5A7C"
source_record_id : "ATFDR-..." (original record being projected)
source_tcs_id : "TCS-..." (TCS embedded at original issuance)
original_record_hash : SHA256(original record JSON at time of projection)
(tamper-evidence commitment to source state)
source_rfv : "EU_AI_ACT_2024_v1.0" (from TCS)
target_rfv : "EU_AI_ACT_2033_v3.0" (current framework at review)
projection_timestamp : "2033-07-14T09:30:00.000000000+00:00"
rulebook_version : "TGB-RAR-V1.0"
rulebook_hash : SHA256(projection_rulebook.json)

field_projections:
- field_name : "counterfactual_outcome"
source_value : "NOMINAL"
target_value : "APPROVED_TIER_A" (mapped per 2033 framework)
projection_rule : "EU_NOMINAL_TO_APPROVED_TIER_A_v1"
projection_confidence: "HIGH"

original_record_integrity: "VERIFIED" | "INVALIDATED" | "UNVERIFIABLE"
invalidation_evidence_id : null | "EID-..." (if INVALIDATED)

SEAL FIELDS:
content_hash_rar : SHA256(canonical JSON of committed fields)
pgc_algorithm : "ML-DSA-65"
rar_seal : base64(ML-DSA-65.sign(content_hash_rar))
```

6.4 Temporal Migration Record (TMR) Specification

A TMR is issued at each evidence lifecycle transition (HOT to WARM, WARM to COLD) as specified in RFC-ATF-3. The TMR captures the regulatory context at the transition moment, providing a timestamped provenance chain for the evidence lifecycle.

```
TMR (Temporal Migration Record) -- identifier: TMR-{16HEX}

COMMITTED FIELDS:
tmr_id : "TMR-8A3F2B1C4D5E6F7A"
source_record_id : "..." (record undergoing lifecycle transition)
transition : "HOT_TO_WARM" | "WARM_TO_COLD"
transition_timestamp : "2026-11-26T00:00:00.000000000+00:00"
pre_transition_lifecycle : "HOT"
post_transition_lifecycle: "WARM"
regulatory_context_at_transition: { same structure as TCS.regulatory_context }
```

```

threshold_context_at_transition: { same structure as TCS.threshold_context }
retention_policy_at_transition: { policy_name, min_retention_years, jurisdiction }

SEAL FIELDS:
content_hash_tmr : SHA256(canonical JSON of committed fields)
pqc_algorithm : "ML-DSA-65"
tmr_seal : base64(ML-DSA-65.sign(content_hash_tmr))

LIFECYCLE:
HOT phase (0 to 6 months): active, fully queryable, high-frequency access
WARM phase (6mo to 3yr): archival, query-accessible, TMR-001 at transition
COLD phase (3yr to 7yr+): long-term retention, retrieval on request, TMR-002 at transition
At 7yr: EU AI Act Art.72 minimum retention satisfied (for high-risk AI)

```

6.5 TGB Projection Rulebook

The TGB projection rulebook is a versioned, PQC-signed document that specifies mapping rules from source framework versions to target framework versions. The rulebook is published by OMNIX QUANTUM and is publicly verifiable. Any party can verify a RAR by obtaining the rulebook and applying the rules to the source TCS.

Rulebook Entry	Source RFV	Target RFV	Mapping Rule	Confidence
EU_NOMINAL_TO_APPROVED_TIER_A	EU_AI_ACT_2024_v1.0	EU_AI_ACT_2027_v2.0	CES >= nominal_threshold_2024 -> APPROVED_TIER_A if threshold preserved	HIGH
EU_HALT_TO_PROHIBITED	EU_AI_ACT_2024_v1.0	EU_AI_ACT_2027_v2.0	outcome == HALT -> SYSTEM_HALT (semantic preservation)	HIGH
NIST_MANAGE22_TO_MANAGE23	NIST_AI_RMF_2023_v1.0	NIST_AI_RMF_2025_v2.0	MANAGE 2.2 provisions -> MANAGE 2.3 equivalent	MEDIUM
THRESHOLD_DELTA_RISK	any	any	If target_threshold != source_threshold -> flag THRESHOLD_CONTEXT_CHANGE	N/A (flag only)

6.6 TGB Invariants: TGB-INV-001-005

Invariant	Statement	Enforcement Mechanism
TGB-INV-001	Every ATF record MUST carry a TCS when TGB_ENABLED=true; TCS MUST be issued and bound synchronously with the primary record	TCS construction synchronous with primary; failure flags TGB_INCOMPLETE on primary record and retries async
TGB-INV-002	RAR MUST NOT modify any field of the source record, source TCS, or any TMR	original_record_hash in RAR provides tamper-evidence commitment; any modification to source detectable from RAR

Invariant	Statement	Enforcement Mechanism
TGB-INV-003	Any party MUST be able to compute a RAR offline using only: source record, TCS, and TGB projection rulebook (PQC-verified)	Rulebook PQC-sealed and publishable; RAR computation requires no platform access or credentials
TGB-INV-004	A framework revision MUST NOT silently invalidate any existing ATF record; any invalidity requires an explicit Evidential Invalidation Document (EID) persisted in RAR table	EID written to <code>atf_regulatory_alignment_receipts</code> with <code>original_record_integrity = INVALIDATED</code>
TGB-INV-005	TMR MUST be issued at each lifecycle transition (HOT->WARM, WARM->COLD) for all records with <code>retention_years >= 7</code> ; TMR MUST carry ML-DSA-65 seal	TMR issuance triggered automatically by lifecycle management module; unsigned TMR rejected at DB insert

6.7 TGB Database Schema

```
-- Table: atf_temporal_context_snapshots (append-only)
CREATE TABLE IF NOT EXISTS atf_temporal_context_snapshots (
  id SERIAL PRIMARY KEY,
  tcs_id VARCHAR(50) UNIQUE NOT NULL,
  parent_record_id VARCHAR(100) NOT NULL,
  parent_record_type VARCHAR(30) NOT NULL,
  issued_at_ns BIGINT NOT NULL,
  issued_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  regulatory_context JSONB NOT NULL,
  threshold_context JSONB NOT NULL,
  deployment_context JSONB,
  tcs_hash VARCHAR(64) NOT NULL,
  pqc_algorithm VARCHAR(30) DEFAULT 'ML-DSA-65',
  tcs_seal TEXT
);

-- Table: atf_regulatory_alignment_receipts (append-only)
CREATE TABLE IF NOT EXISTS atf_regulatory_alignment_receipts (
  id SERIAL PRIMARY KEY,
  rar_id VARCHAR(50) UNIQUE NOT NULL,
  source_record_id VARCHAR(100) NOT NULL,
  source_tcs_id VARCHAR(50) NOT NULL,
  original_record_hash VARCHAR(64) NOT NULL,
  source_rfv VARCHAR(100) NOT NULL,
  target_rfv VARCHAR(100) NOT NULL,
  projection_timestamp TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  rulebook_version VARCHAR(30) NOT NULL,
  rulebook_hash VARCHAR(64),
  field_projections JSONB,
```

```

original_record_integrity VARCHAR(20) NOT NULL DEFAULT 'VERIFIED'
CHECK (original_record_integrity IN ('VERIFIED','INVALIDATED','UNVERIFIABLE')),
invalidation_evidence_id VARCHAR(50),
content_hash_rar VARCHAR(64) NOT NULL,
pqc_algorithm VARCHAR(30) DEFAULT 'ML-DSA-65',
rar_seal TEXT
);

-- Table: atf_temporal_migration_records
CREATE TABLE IF NOT EXISTS atf_temporal_migration_records (
id SERIAL PRIMARY KEY,
tmr_id VARCHAR(50) UNIQUE NOT NULL,
source_record_id VARCHAR(100) NOT NULL,
transition VARCHAR(20) NOT NULL
CHECK (transition IN ('HOT_TO_WARM','WARM_TO_COLD')),
transition_timestamp TIMESTAMPTZ NOT NULL DEFAULT NOW(),
pre_transition_lifecycle VARCHAR(10) NOT NULL,
post_transition_lifecycle VARCHAR(10) NOT NULL,
regulatory_context_at_transition JSONB NOT NULL,
threshold_context_at_transition JSONB,
retention_policy_at_transition JSONB,
content_hash_tmr VARCHAR(64) NOT NULL,
pqc_algorithm VARCHAR(30) DEFAULT 'ML-DSA-65',
tmr_seal TEXT
);

```

7. Formal Verification (OMNIX-FVS-1.0 Extension)

RFC-ATF-5 extends the OMNIX Formal Verification Suite (FVS-1.0) defined in RFC-ATF-4. RFC-ATF-5 maintains the dual methodology: Z3 SMT for arithmetic and structural invariants, TLA+ for state-machine safety and liveness. This is the only AI governance standard with machine-checkable formal proofs across both methodologies.

7.1 Z3 SMT Proof Targets

Target ID	Module	Property Proven	Z3 Result
CGE-FRAGILITY-BOUND-LO	CGE	fragility_score >= 0.0 for all valid real inputs	UNSAT
CGE-FRAGILITY-BOUND-HI	CGE	fragility_score <= 1.0 for all valid real inputs	UNSAT
CGE-VV-BOUND	CGE	abs(VV.delta) <= CGE_MAX_VARIATION_PCT for all VV fields	UNSAT
CGE-PRIMARY-ISOLATION	CGE	CGE computation cannot modify any committed field of primary record	UNSAT
CGE-ROOT-HASH-INTEGRITY	CGE	cat_root_hash uniquely determined by sorted(cfr_content_hashes)	UNSAT
GUGT-UGI-COMPLETENESS	GUGT	GUGT_COMPLIANT iff all 6 UGIs are PASS (no fewer, no more)	UNSAT
GUGT-UGI-MONOTONICITY	GUGT	GUGT-Lk satisfaction logically implies GUGT-Lj for all j < k	UNSAT
GUGT-WAIVER-PROHIBITION	GUGT	No UIR can satisfy GUGT invariants while waiving any ATF invariant	UNSAT
TGB-NON-DESTRUCTION	TGB	RAR production cannot change original_record_hash or source TCS fields	UNSAT
TGB-PROJECTION-MONOTONE	TGB	Framework revision cannot silently increase compliance level	UNSAT
TGB-FRAGILITY-PRESERVATION	TGB	fragility_score value in CAT preserved unchanged by TCS/RAR operations	UNSAT
ATF-INV-001-CGL-EXT	Inherited	MAR invariant preserved under all CGE/GUGT/TGB operations	UNSAT

7.2 TLA+ Coverage

The TLA+ specification for RFC-ATF-5 extends the prior ATF TLA+ model (ATF-TLA-SPEC.tla) with three new modules: CGE_MODULE.tla, GUGT_MODULE.tla, and TGB_MODULE.tla. Each module specifies state transitions, safety properties, and liveness conditions for the respective CGL component.

TLA+ Specification	Property Type	Coverage	Status
CGE_MODULE.tla -- CGE_ISOLATION	Safety	Primary record state unchanged by CGE transitions	PASS

TLA+ Specification	Property Type	Coverage	Status
CGE_MODULE.tla -- CAT_EVENTUALLY_PRODUCED	Liveness	CAT is eventually produced for every evaluation with CGE_ENABLED	PASS
GUGT_MODULE.tla -- UIR_CONSISTENCY	Safety	UIR overall_status consistent with UGI status set at all states	PASS
GUGT_MODULE.tla -- LEVEL_HIERARCHY	Safety	Conformance level hierarchy monotonicity preserved across all transitions	PASS
TGB_MODULE.tla -- TCS_IMMUTABILITY	Safety	TCS fields cannot be modified after persist transition	PASS
TGB_MODULE.tla -- RAR_NON_DESTRUCTION	Safety	Source record state unchanged by RAR production transition	PASS
TGB_MODULE.tla -- TMR_AT_TRANSITION	Liveness	TMR is eventually produced at every lifecycle transition for retained records	PASS
ATF_INTEGRATION.tla -- CGL_NON_BLOCKING	Safety	Primary record state ACTIVE regardless of CGL module failure states	PASS

8. Combined Invariant Summary -- 88 Total

RFC-ATF-5 introduces 18 new invariants across three protocol families. Combined with the 70 invariants from RFC-ATF-1 through RFC-ATF-4, the complete ATF stack encompasses **88 formally specified invariants** across 17 protocol families.

Family	RFC	Count	Scope
ATF-INV	RFC-ATF-1	6	Identity & Delegation -- MAR, Acyclicity, Chain Root, Immutability, Non-Future-Dating, Offline Verifiability
RGC-INV	RFC-ATF-2	8	Runtime Continuity -- CES, AFG, HALT propagation, RC, RCR integrity
GPIL-INV	RFC-ATF-3	3	Governance Policy Interoperability Layer
ELR-INV	RFC-ATF-3	4	Evidence Lifecycle Record management
EAP-INV	RFC-ATF-3	7	Evidence Archive Pipeline
OEP-INV	RFC-ATF-3	6	OMNIX Evidence Package -- two-phase PQC seal, offline verification
FEA-INV	RFC-ATF-3	5	Forensic Export Authorization -- caller-key controls
FVP-INV	RFC-ATF-3	1	Forensic Verification Protocol
GECD-INV	RFC-ATF-3	6	Governance Execution Context Record
SGIP-INV	RFC-ATF-3	4	Semantic Governance Interoperability Protocol
DSPP-INV	RFC-ATF-4	7	Dynamic Semantic Portability Protocol -- TSA, SDR, RSA
AGV-INV	RFC-ATF-4	6	Anticipatory Governance Veto Protocol -- PVR
SSD-INV	RFC-ATF-4	3	Structural Shift Detection -- CRSI
FVS-INV	RFC-ATF-4	3	Formal Verification Suite extension
CGE-INV	RFC-ATF-5	7	Counterfactual Governance Engine [NEW]
GUGT-INV	RFC-ATF-5	6	Grand Unified Governance Theory [NEW]
TGB-INV	RFC-ATF-5	5	Temporal Governance Bridge [NEW]
TOTAL	--	88	Complete ATF Protocol Stack, five RFCs, 17 families

9. ATF-CGL-Compliant Designation

ATF COMPLIANCE HIERARCHY (five tiers, strictly hierarchical):

```
+-----+
| * ATF-CGL-Compliant (RFC-ATF-1+2+3+4+5) 88 invariants HIGHEST|
| All five RFC layers operational; CGE + GUGT UIR + TGB required |
+-----+
```

^--- extends without replacing

```
+-----+
| ATF-PGL-Compliant (RFC-ATF-1+2+3+4) 70 invariants |
| AGVP + SSD + DSPP operational |
+-----+
```

^--- extends without replacing

```
+-----+
| ATF-FEI-Compliant (RFC-ATF-1+2+3) 40 invariants |
| OEP + GPIL + Forensic verification operational |
+-----+
```

^--- extends without replacing

```
+-----+
| ATF-RGC-Compliant (RFC-ATF-1+2) 14 invariants |
| Runtime continuity + HALT propagation operational |
+-----+
```

^--- extends without replacing

```
+-----+
| ATF-Compliant-L1/2/3 (RFC-ATF-1) 6 invariants |
| Identity + Delegation + Trust Lattice operational |
+-----+
```

ATF-CGL-Compliant requirements (all five required simultaneously):

- (a) ATF-PGL-Compliant: all RFC-ATF-1/2/3/4 requirements satisfied
- (b) CGE operational: CGE_ENABLED=true, CGE_FORK_COUNT in [1,7], all CGE-INV-001-007 satisfied, CATs produced for all evaluations
- (c) GUGT UIR issued: at least one GUGT-L3+ATF UIR issued, PQC-sealed, and verified for the deployment
- (d) TGB operational: TGB_ENABLED=true, all records carry TCS, all TGB-INV-001-005 satisfied
- (e) CGL formal verification: FVS-1.0 extension runs all CGL Z3 proof targets returning UNSAT; TLA+ modules pass all safety/liveness checks

10. Implementation Reference

10.1 Reference Implementation Components

Component	Role	Public Interface
CGE Engine	CFR computation, VV generation, CAT assembly, fragility_score computation, CGE-INV enforcement	/api/atf/cge/evaluate -- /api/atf/cge/cat/{evaluation_id}
GUGT Assessor	UGI-001-006 evaluation, UIR issuance, conformance level assignment, GUGT-INV enforcement	/api/atf/gugt/assess -- /api/atf/gugt/uir/{deployment_id}
TGB Manager	TCS embedding, RAR production, TMR at lifecycle transitions, rulebook management	/api/atf/tgb/tcs/{record_id} -- /api/atf/tgb/rar
CGE Verifier	Offline CAT verification: root hash, CFR signatures, VV determinism, fragility_score	python omnix_atf_verify.py --cat cat.json
GUGT Verifier	Offline UIR verification: UGI status set, ML-DSA-65 seal, conformance level	python omnix_atf_verify.py --uir uir.json
TGB Projector	Offline RAR computation: TCS + rulebook + source record -> RAR projection	python omnix_tgb_project.py --tcs tcs.json --record record.json --rulebook rulebook.json
FVS Runner	Z3 + TLA+ proof execution: all CGL proof targets, result reporting	python omnix_fvs_run.py --suite cgl -- outputs: UNSAT for all Z3, PASS for all TLA+
CGL DB Manager	6 new table creation, append-only enforcement triggers, FK integrity	Auto-created on first request via CREATE TABLE IF NOT EXISTS

10.2 API Endpoints

Endpoint	Method	Description	Auth
/api/atf/cge/evaluate	POST	Trigger CGE evaluation for a primary receipt; returns cat_id	B2B API key
/api/atf/cge/cat/{evaluation_id}	GET	Retrieve CAT and all CFRs for an evaluation	B2B API key
/api/atf/cge/verify	POST	Online CAT verification (offline CLI also available)	B2B API key
/api/atf/gugt/assess	POST	Run GUGT assessment for a deployment; returns uir_id	Admin
/api/atf/gugt/uir/{deployment_id}	GET	Retrieve most recent UIR for a deployment	B2B API key
/api/atf/tgb/tcs/{record_id}	GET	Retrieve TCS for a specific ATF record	B2B API key
/api/atf/tgb/rar	POST	Request RAR production for a historical record	Admin
/api/atf/tgb/project	POST	Offline-compatible RAR projection request	Admin

10.3 Configuration Parameters

Parameter	Default	Range	Description
CGE_ENABLED	false	true/false	Enable CGE counterfactual computation
CGE_FORK_COUNT	3	1-7	Number of CFRs to compute per evaluation
CGE_MAX_VARIATION_PCT	0.40	0.05-0.40	Maximum absolute VV field delta allowed
CGE_ASYNC_TIMEOUT_S	30	5-300	Timeout for async CGE computation
TGB_ENABLED	false	true/false	Enable TCS embedding and TGB operations
TGB_SYNC_TCS	true	true/false	Embed TCS synchronously (false = async, less safe)
TGB_RETENTION_YEARS	7	1-50	Default evidence retention period for TMR triggers
TGB_RULEBOOK_URL	(internal)	URL	URL of signed TGB projection rulebook
GUGT_AUTO_ASSESS	false	true/false	Auto-run GUGT assessment on deployment startup
GUGT_CONFORMANCE_TARGET	GUGT-L3+ATF	L1/L2/L3/L3+ATF	Minimum conformance level for deployment health check

11. Security Considerations

Threat	Attack Vector	CGL Mitigation	Residual Risk
CGE Primary Manipulation (CGL-T-001)	Attacker attempts to modify primary record outcome via CGE computation path	CGE-INV-002: CGE is strictly read-only post-primary-commit; Z3 proof CGE-PRIMARY-ISOLATION	NONE -- structural guarantee
CFR Fabrication (CGL-T-002)	Attacker injects fabricated CFRs to distort fragility_score or decision space	CGE-INV-003: all CFRs require valid ML-DSA-65 signature; CGE-INV-007: VV determinism verifiable offline	LOW -- requires key compromise
CAT Root Hash Collision (CGL-T-003)	Attacker replaces CFRs after CAT issued, with colliding root hash	SHA-256 preimage resistance; cat_root_hash binding via cat_seal ML-DSA-65	NEGLIGIBLE -- SHA-256 + PQC
UIR Level Inflation (CGL-T-004)	Attacker claims GUGT-L3+ATF without meeting all requirements	GUGT-INV-006: level monotonicity enforced; Z3 proof GUGT-UGI-MONOTONICITY	LOW -- requires UIR key compromise
TCS Retroactive Modification (CGL-T-005)	Attacker modifies TCS to change historical context	TGB-INV-002: tcs_hash bound in parent record posture_state_hash; Z3 proof TGB-NON-DESTRUCTION	NONE -- tamper-evident from primary
Temporal Downgrade (CGL-T-006)	Attacker provides outdated rulebook to produce favorable RAR projection	TGB-INV-003: rulebook PQC-sealed and versioned; rulebook_hash in RAR	LOW -- requires rulebook key compromise
Silent Framework Invalidation (CGL-T-007)	Framework revision used to claim prior records non-compliant without notice	TGB-INV-004: EID required for any invalidity; silent invalidity rejected	NONE -- explicit EID required
UGI Waiver Attack (CGL-T-008)	UIR claims to waive an ATF invariant via GUGT assessment override	GUGT-INV-005: waiver of any ATF invariant causes UIR rejection; Z3 proof GUGT-WAIVER-PROHIBITION	NONE -- structural guarantee

12. Novel Contributions

1. Counterfactual Attestation Token (CAT) with PQC Seal

First governance artifact that cryptographically seals the decision space (not just the selected path) using post-quantum signatures. Existing counterfactual analysis tools (IBM OpenScale, Microsoft Responsible AI Toolbox) produce reports, not tamper-evident artifacts. The CAT is independently verifiable offline by any party holding only the receipt artifacts and the public key.

2. fragility_score -- First Governance Decision Robustness Metric

The fragility_score quantifies governance decision robustness as a continuous metric in $[0.0, 1.0]$, formally defined as $\text{divergence_count} / \text{cfr_count}$. No published AI governance specification defines a quantitative robustness metric for governance decisions. The fragility_score is PQC-sealed within the CAT and independently reproducible from the CFR set.

3. Universal Governance Invariants (UGI-001-006)

First formally derived minimal set of universal governance requirements obtained by exhaustive clause-level intersection of five major regulatory frameworks simultaneously. Prior frameworks (EU AI Act guidance, NIST AI RMF profiles) are jurisdiction-specific. No published specification derives a minimal universal set via formal intersection methodology.

4. Universal Invariant Receipt (UIR) -- Cross-Framework PQC Certificate

First PQC-signed artifact that certifies compliance with five regulatory frameworks and six agent types simultaneously in a single verifiable record. Existing compliance certification systems are framework-specific and platform-dependent. The UIR is offline-verifiable and architecture-agnostic.

5. Temporal Context Snapshot (TCS) -- Nanosecond-Precision Regulatory Context Capture

First record type designed to capture the complete regulatory and threshold context at nanosecond precision at the moment of record creation. Prior governance specifications embed timestamps but not the interpretive framework under which the timestamp was produced. Without TCS, a historical receipt is a fact without context. With TCS, it is a self-interpreting artifact.

6. Regulatory Alignment Receipt (RAR) -- Non-Destructive Temporal Projection

First mechanism for projecting historical governance records to current regulatory frameworks without modifying the original evidence. The non-destruction invariant (TGB-INV-002) is formally proved via Z3 (TGB-NON-DESTRUCTION). No prior AI governance specification provides a non-destructive, offline-computable, PQC-signed temporal projection mechanism.

7. ATF-CGL-Compliant -- Five-Tier Formally Verified Compliance Hierarchy

First five-tier AI governance compliance designation backed by 88 formally specified invariants across 17 protocol families, dual formal verification (Z3 + TLA+), and post-quantum cryptography. No published AI governance standard defines a multi-tier compliance hierarchy with machine-checkable invariants across identity, runtime, evidence, proactive, and cognitive governance dimensions.

13. Distinction from Prior Art

Feature	RFC-ATF-5 (CGL)	VeriSigil VGS	IBM OpenScale	MS Azure RAI	NIST AI RMF
Decision space evidence (PQC-sealed)	[OK] CAT -- offline-verifiable CFRs	[NO]	[NO]	[NO]	[NO]
fragility_score metric	[OK] Defined, Z3-proved in [0,1]	[NO]	[NO]	[NO]	[NO]
Universal governance invariants	[OK] UGI-001-006 by intersection	[NO]	[NO]	[NO]	Partial (single-framework)
PQC-signed universal cert (UIR)	[OK] ML-DSA-65	[NO]	[NO]	[NO]	[NO]
Multi-jurisdiction UIR	[OK] EU+US+GCC+ISO+UK	Partial (EU only)	[NO]	Partial	[NO]
TCS at record issuance	[OK] Nanosecond precision	[NO]	[NO]	[NO]	[NO]
Non-destructive RAR	[OK] TGB-INV-002 (Z3-proved)	[NO]	[NO]	[NO]	[NO]
7-year retention support (TMR)	[OK] HOT/WARM/COLD + PQC	[NO]	[NO]	[NO]	[NO]
Dual Z3 + TLA+ formal proofs	[OK] 12 Z3 targets + 8 TLA+	[OK] Z3 only	[NO]	[NO]	[NO]
Offline full-chain verification	[OK] CLI -- no platform	[NO]	[NO]	[NO]	[NO]
Post-quantum cryptography	[OK] ML-DSA-65 FIPS 204	[NO]	[NO]	[NO]	[NO]

14. Regulatory Alignment

Framework / Article	CGE Contribution	GUGT Contribution	TGB Contribution
EU AI Act Art. 9 -- Risk Management	"Alternatives considered" documentation: CAT satisfies with PQC-signed artifacts, not narrative	"Risk management system" completeness: GUGT-L3+ATF certifies Art.9 complete by construction	Art.9(7) assessment integrity: TCS captures assessment context at issuance
EU AI Act Art. 11-12 -- Technical Doc	CFR set provides full decision space documentation for technical file	UIR provides single artifact covering Art.11 requirements across systems	TCS provides issuance-time context for technical documentation accuracy
EU AI Act Art. 72 -- Record Keeping	CAT provides decision space records for 7-year retention	UIR provides single compliance artifact for multi-year retention	TMR at lifecycle transitions + RAR at review time: 7-year bridge
NIST AI RMF MAP 1.6 -- Impact Analysis	Alternative scenario documentation via CFR set	MAP 5.1/5.2 complete coverage: UIR certifies by construction	MAP 5.2 lifecycle documentation: TMR at each stage
GCC/DIFC Art. 8-14 -- AI Governance	Arts. 8-14 "alternatives assessment": CAT covers	Complete UIR coverage of all GCC/DIFC requirements	Art.12/14 audit trail across framework revisions via RAR
ISO/IEC 42001 Sec.6-9 -- AI MS	Sec.6.1.2 risk treatment alternatives: CFRs	Sec.6.1.2/8.4/8.5/9.3 complete: UIR certifies full coverage	Sec.9.1/9.3 management review with historical context via TCS
SOC 2 Type II CC7.2 -- Monitoring	Decision space documentation for change monitoring	Evidence trail satisfying monitoring control requirements	Change documentation at each lifecycle transition via TMR
UK AISI -- Evaluation Framework	Sec.3-5 "alternative outcomes considered" requirement	All AISI evaluation criteria satisfied: GUGT-L3+ATF includes UK AISI	Evaluation evidence preserved across temporal transitions

15. Known Limitations and Open Questions

Limitation	Mitigation	ADR Reference	Status
CGE computation latency for M=7 forks on complex governance pipelines	CGE async by design; timeout configurable via CGE_ASYNC_TIMEOUT_S; partial CAT accepted (CGE-INV-001)	ADR-178	MANAGED
Rulebook coverage gaps: not all source/target RFV pairs may have projection rules	Rulebook versioned and extensible; gap flagged as UNVERIFIABLE in RAR.original_record_integrity	ADR-180	OPEN
TCS synchronous embedding adds ~100ns to record creation latency	Non-blocking failure mode (TGB_INCOMPLETE flag + async retry); TGB_SYNC_TCS=false for latency-critical deployments	ADR-180	MANAGED
UIR is per-deployment, not per-record; deployment configuration changes may require UIR refresh	GUGT_AUTO_ASSESS=true triggers re-assessment on configuration change; prior UIRs are historical artifacts	ADR-179	MANAGED
VV space exploration is parametric, not exhaustive; true decision space may be larger	fragility_score communicates this limitation explicitly; CGE_FORK_COUNT up to 7 for higher coverage	ADR-178	BY DESIGN
RAR projection requires trusted rulebook; rulebook unavailability blocks offline RAR	Rulebook PQC-sealed and publishable independently; local cache recommended for offline environments	ADR-180	OPEN
Multi-instance deployments require shared TCS signing key or key synchronization	Covered by existing OMNIX signing key management; no CGL-specific issue	ADR-178/179/180	INHERITED

16. References

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- [**ADR-178**] Nunes, H., "ADR-178: Counterfactual Governance Engine (CGE)", OMNIX QUANTUM Architecture Decision Record, May 2026.
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- [**W3CVC**] W3C, "Verifiable Credentials Data Model 2.0", W3C Recommendation, May 2024.
- [**CRYSTALS**] Ducas, L., et al., "CRYSTALS-Dilithium: A Lattice-Based Digital Signature Scheme", IACR TCHES 2018(1), 2018.

Appendix A -- CGL Wire Formats (JSON Schemas)

A.1 CFR Wire Format

```
{
  "$schema": "https://omnixquantum.net/schemas/atf/cgl/cfr-v1.0.json",
  "cfr_id": "CFR-[16 uppercase hex]",
  "cat_id": "CAT-[16 uppercase hex]",
  "primary_receipt_id": "[parent record ID]",
  "evaluation_id": "CGE-EVAL-[16 uppercase hex]",
  "variation_vector": { "[field_name]": [numeric_delta], ... },
  "counterfactual_ces_score": [0.0 - 100.0],
  "counterfactual_outcome": "NOMINAL|MONITORING|WARNING|HALT",
  "diverges_from_primary": true | false,
  "cfr_status": "ACCEPTED|REJECTED",
  "rejection_reason": null | "CGE_VV_BOUND_VIOLATION",
  "vv_seed": "[64-char hex SHA256(eval_id || receipt_id)]",
  "posture_state_hash_cfr": "[64-char hex SHA256 of committed fields]",
  "content_hash_cfr": "[64-char hex SHA256 of all fields excl. sig]",
  "pqc_algorithm": "ML-DSA-65",
  "pqc_signature": "[base64 ML-DSA-65 signature over content_hash_cfr]",
  "issued_at": "[ISO8601 with nanoseconds and timezone]"
}
```

A.2 CAT Wire Format

```
{
  "$schema": "https://omnixquantum.net/schemas/atf/cgl/cat-v1.0.json",
  "cat_id": "CAT-[16 uppercase hex]",
  "primary_receipt_id": "[parent record ID]",
  "evaluation_id": "CGE-EVAL-[16 uppercase hex]",
  "cfr_ids": ["CFR-...", "CFR-...", ...],
  "cfr_count": [integer 1-7],
  "divergence_count": [integer 0 - cfr_count],
  "cat_root_hash": "[sha256(sorted_cfr_hashes_joined_by_pipe)]",
  "fragility_score": [decimal 0.0000 - 1.0000],
  "cge_fork_count_config": [integer 1-7],
  "content_hash_cat": "[64-char hex SHA256 of all fields excl. seal]",
  "pqc_algorithm": "ML-DSA-65",
  "cat_seal": "[base64 ML-DSA-65 signature over content_hash_cat]",
  "issued_at": "[ISO8601 with nanoseconds and timezone]"
}
```

A.3 UIR Wire Format

```
{
  "$schema": "https://omnixquantum.net/schemas/atf/cgl/uir-v1.0.json",
  "uir_id": "UIR-[16 uppercase hex]",
  "assessed_protocol": "ATF",
  "atf_spec_version": "RFC-ATF-5_v1.0",
  "agent_type_coverage": ["LLM", "FINANCIAL", "MEDICAL", "ROBOTIC", ...],
  "ugi_001_status": "PASS|PARTIAL|FAIL",
  "ugi_001_evidence_ref": "[ATF record ID used as evidence]",
  "ugi_002_status": "PASS|PARTIAL|FAIL",
  "ugi_002_evidence_ref": "[ATF record ID]",
  "ugi_003_status": "PASS|PARTIAL|FAIL",
  "ugi_003_evidence_ref": "[ATF record ID]",
  "ugi_004_status": "PASS|PARTIAL|FAIL",
  "ugi_004_evidence_ref": "[ATF record ID]",
  "ugi_005_status": "PASS|PARTIAL|FAIL",
  "ugi_005_evidence_ref": "[ATF record ID]",
  "ugi_006_status": "PASS|PARTIAL|FAIL",
  "ugi_006_evidence_ref": "[ATF record ID]",
  "overall_gugt_status": "GUGT_COMPLIANT|GUGT_PARTIAL|GUGT_NON_COMPLIANT",
  "conformance_level": "GUGT-L1|GUGT-L2|GUGT-L3|GUGT-L3+ATF",
  "framework_coverage": ["EU_AI_ACT_ART14", "NIST_GOVERN_1.1", ...],
  "content_hash_uir": "[64-char hex SHA256 of all fields excl. seal]",
  "pqc_algorithm": "ML-DSA-65",
  "uir_seal": "[base64 ML-DSA-65 signature over content_hash_uir]",
  "issued_at": "[ISO8601 with nanoseconds and timezone]"
}
```

A.4 TCS Wire Format

```
{
  "$schema": "https://omnixquantum.net/schemas/atf/cgl/tcs-v1.0.json",
  "tcs_id": "TCS-[16 uppercase hex]",
  "parent_record_id": "[ATF record ID this TCS is embedded in]",
  "parent_record_type": "DR|TAR|RCR|GovernanceReceipt|...",
  "issued_at_ns": [integer UNIX nanoseconds],
  "issued_at": "[ISO8601 with nanoseconds and timezone]",
  "regulatory_context": {
    "eu_ai_act_version": "[string]",
    "nist_ai_rmf_version": "[string]",
    "gcc_difc_version": "[string]",
    "iso_42001_version": "[string]",
    "atf_spec_version": "[string]",
  }
}
```

```
"active_adr_count": [integer]
},
"threshold_context": {
  "nominal_threshold": [decimal],
  "monitoring_lower": [decimal],
  "warning_lower": [decimal],
  "halt_threshold": [decimal],
  "fragmentation_limit": [decimal],
  "max_delegation_depth": [integer]
},
"tcs_hash": "[SHA256 of committed fields]",
"pqc_algorithm": "ML-DSA-65",
"tcs_seal": "[base64 ML-DSA-65 signature over tcs_hash]"
}
```

Appendix B -- GUGT Framework Clause Mapping

The following table provides the complete clause-level mapping used to derive UGI-001-006 from the five regulatory frameworks. Each cell identifies the specific clause(s) that require the corresponding universal property.

Universal Property	EU AI Act 2024	NIST AI RMF 1.0	GCC/DIFC 2024	ISO/IEC 42001:2023	UK AISI Eval Fwk
Human authority anchor to identified principal	Art.14(4): human oversight; Art.17: governance system with identified persons	GOVERN 1.1: roles, responsibilities, and authorities defined	Art.8: human oversight requirements	Sec.6.2: objectives and planning; Sec.8.1: operational planning	Sec.3: responsible parties and accountability
Offline-verifiable decision evidence	Art.11-12: technical documentation and record-keeping accessible to authorities	MAP 5.2: AI system testing and evaluation documentation	Art.12: documentation and audit trails available to authorities	Sec.9.1: monitoring, measurement, analysis -- evidence requirements	Sec.4: evidence standards for evaluation
Execution-time boundary enforcement	Art.9: risk management system -- controls at point of operation	MANAGE 2.2: mechanisms for AI response to undesirable behavior	Art.9: operational controls at deployment	Sec.8.4: operational planning and control	Sec.4: runtime enforcement requirements
Pre-committed posture assessment	Art.9(7): risk management documentation before operation	MEASURE 2.5: AI system performance evaluation before deployment	Art.9: pre-deployment assessment requirements	Sec.8.5: objectives implementation -- pre-committed documentation	Sec.4: assessment methodology requirements
Self-modification prohibition	Art.14(4): human oversight maintains ability to override	GOVERN 6.1: policies regarding AI modification	Art.10: control of AI system modifications	Sec.6.1.2: risk assessment for AI modifications	Sec.5: control requirements for AI behavior modification
Self-contained evidence reconstruction	Art.18: post-market monitoring -- complete evidence without platform	MAP 5.1: AI system impact assessment documentation	Art.14: complete documentation available for audit	Sec.9.3: management review with complete evidence	Sec.5: evidence sufficiency for independent review

Appendix C -- ATF-CGL-Compliant Checklist

The following checklist provides the complete set of requirements for achieving ATF-CGL-Compliant designation. All items are required.

C.1 Prior RFC Compliance (ATF-PGL-Compliant required)

- [] ATF-Compliant-L3: AIR, DR, TAR, Trust Lattice operational; ATF-INV-001-006 all satisfied
- [] ATF-RGC-Compliant: RCR, CES, AFG, RC operational; RGC-INV-001-008 all satisfied
- [] ATF-FEI-Compliant: OEP, GPIL, ELC, FVP operational; all L3 invariants satisfied
- [] ATF-PGL-Compliant: AGVP, SSD, DSPP operational; all L4 invariants satisfied

C.2 CGE Requirements

- [] CGE_ENABLED=true in production configuration
- [] CGE_FORK_COUNT in [1, 7]; CGE_MAX_VARIATION_PCT <= 0.40
- [] CGE-INV-001: CAT produced for every evaluation; CGE_INCOMPLETE flag operational
- [] CGE-INV-002: Primary record isolation verified; CGE reads primary post-commit only
- [] CGE-INV-003: All CFRs carry valid ML-DSA-65 signature; unsigned CFRs rejected
- [] CGE-INV-004: cat_root_hash computation correct; fragility_score in [0.0, 1.0]
- [] CGE-INV-005: VV bound enforcement operational; violations logged
- [] CGE-INV-006: DB triggers enforce append-only on counterfactual tables
- [] CGE-INV-007: Offline verification procedure documented and tested
- [] Z3 proof targets CGE-FRAGILITY-BOUND-LO/HI, CGE-VV-BOUND, CGE-PRIMARY-ISOLATION: UNSAT

C.3 GUGT Requirements

- [] GUGT assessment conducted for deployment; UIR issued with GUGT-L3+ATF status
- [] UIR covers all 6 UGIs; overall_gugt_status = GUGT_COMPLIANT
- [] UIR carries valid ML-DSA-65 seal; GUGT-INV-003 satisfied
- [] agent_type_coverage explicitly declared; GUGT-INV-004 satisfied
- [] framework_coverage uses specific clause references (FRAMEWORK_CLAUSE format)
- [] GUGT-INV-005: no ATF invariant waived; Z3 proof GUGT-WAIVER-PROHIBITION: UNSAT
- [] GUGT-INV-006: conformance level hierarchy monotonic; Z3 proof GUGT-UGI-MONOTONICITY: UNSAT

C.4 TGB Requirements

- [] TGB_ENABLED=true in production configuration
- [] TGB-INV-001: TCS embedded in every ATF record; TGB_INCOMPLETE flag operational

- [] TCS carries valid ML-DSA-65 seal; tcs_hash bound in parent record posture_state_hash
- [] TGB-INV-002: RAR production does not modify source record; Z3 proof TGB-NON-DESTRUCTION:
UNSAT
- [] TGB-INV-003: Offline RAR computation procedure documented; rulebook PQC-sealed
- [] TGB-INV-004: Framework revision scan operational; EID issued on invalidity
- [] TGB-INV-005: TMR issued at HOT->WARM and WARM->COLD transitions for retained records

C.5 Formal Verification

- [] All 12 Z3 proof targets return UNSAT (see Section 7.1)
- [] All 8 TLA+ specification checks pass (see Section 7.2)
- [] FVS runner output archived with UIR as supporting evidence

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DRAFT -- pending Zenodo submission